

Northern University of Business and Technology, Khulna

Department of Computer Science and Engineering (CSE)



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Project Title: AI Study Assistant Platform

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1.INTRODUCTION

1.1 Project Overview

AI Study Assistant Platform is an AI-powered educational web application designed to help students improve learning efficiency through intelligent automation and academic assistance.

The system provides several smart features such as AI chatbot support, smart note summarization, quiz generation, task management, and personalized study planning.

The project combines modern full-stack web technologies with Artificial Intelligence to create a more interactive and productive learning environment for students.

1.2 Objectives

The main objectives of the project are:

- Build an AI-powered study assistant platform
- Improve student productivity and study management
- Provide intelligent academic support using AI chatbot
- Generate quizzes automatically from topics
- Summarize study notes using Artificial Intelligence
- Create personalized study plans
- Develop a responsive and user-friendly web application

1.3 Scope

The system will provide a web-based platform where students can manage study activities, communicate with an AI chatbot, organize tasks, generate quizzes, summarize notes, and receive personalized learning assistance.

The system will mainly focus on:

- Student productivity improvement
- Intelligent academic assistance
- Study planning and organization
- AI-powered automation

The project will be developed as a responsive full-stack web application.

2. FUNCTIONAL REQUIRMENTS

Functional requirements describe the functionalities and operations of the system.

2.1 User Authentication

- Users can register accounts
- Users can login/logout securely
- Users can access personalized dashboards
- Passwords will be encrypted
- Users can update profile information

2.2 AI Chatbot System

- Users can ask academic questions
- AI generates intelligent responses
- Chat history can be stored
- AI can explain educational topics
- AI provides learning assistance

2.3 Study Planner System

- Users can create study tasks
- Users can update tasks
- Users can delete tasks
- Users can mark tasks as completed
- Users can organize study schedules

2.4 Quiz Generation System

- Users can enter quiz topics
- AI generates quiz questions automatically
- Users can attempt quizzes
- Quiz history can be stored

2.5 Notes Summarization System

- Users can upload notes or text
- AI generates summarized content
- Users can save summaries
- Users can review summarized notes later

2.6 Dashboard System

- Users can view study progress
- Users can track completed tasks
- Users can view quiz performance
- Dashboard displays personalized information

3. NON- FUNCTIONAL REQUIRMENTS

Non-functional requirements describe how the system should behave.

3.1 Performance Requirements

- Fast API response time
- Efficient database queries
- Smooth user experience
- Quick AI response generation

3.2 Security Requirements

- JWT authentication
- Password hashing using bcrypt
- Secure API communication
- Protected user data
- Environment variable protection

3.3 Usability Requirements

- Responsive user interface
- Easy navigation system
- User-friendly dashboard
- Mobile-friendly design

3.4 Reliability Requirements

- Stable backend server
- Error handling support
- Reliable database connection
- System availability during usage

3.5 Scalability Requirements

- Future feature expansion support
- Scalable database structure
- Modular backend architecture

4. USER ROLES

Role	Description
Student	Main user who accesses study tools and AI features
Admin	Manages users and monitors system activities

5. USE CASES

5.1 User Registration

Actor

Student

Action

User enters registration information

System Response

System validates data and creates user account

5.2 User Login**Actor**

Student

Action

User enters email and password

System Response

System verifies credentials and logs in user

5.3 AI Chatbot Interaction**Actor**

Student

Action

User asks academic question

System Response

AI generates intelligent response

5.4 Quiz Generation**Actor**

Student

Action

User enters quiz topic

System Response

AI generates quiz questions

5.5 Task Management

Actor

Student

Action

User creates or updates study task

System Response

Task is saved and displayed on dashboard

5.6 Notes Summarization

Actor

Student

Action

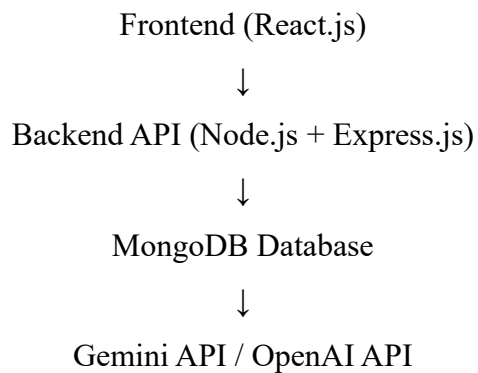
User uploads notes or text

System Response

AI generates summarized notes

6. SYSTEM ARCHITECTURE

The system architecture follows a modern full-stack structure.



7. TECHNOLOGY STACK

Component	Technology
Frontend	React.js, Tailwind CSS
Backend	Node.js, Express.js
Database	MongoDB Atlas
Authentication	JWT, bcrypt
AI Integration	Gemini API / OpenAI API
Deployment	Vercel, Render
Version Control	Git & GitHub

8. DATABASE DESIGN OVERVIEW

The proposed database will contain the following collections/tables:

- Users
- Tasks
- Notes
- ChatHistory
- Quizzes

8.1 Users Collection

Field	Type
_id	ObjectId
name	String
email	String
password	String

8.2 Tasks Collection

Field	Type
_id	ObjectId
userId	ObjectId
title	String
status	String

8.3 Notes Collection

Field	Type
_id	ObjectId
userId	ObjectId
content	String
summary	String

8.4 ChatHistory Collection

Field	Type
_id	ObjectId
userId	ObjectId
userMessage	String
aiResponse	String
createdAt	Date

8.5 Quizzes Collection

Field	Type
_id	ObjectId
userId	ObjectId
topic	String
questions	Array
score	Number
createdAt	Date

9. FUTURE SCOPE

Future improvements of the system may include:

- Mobile application development
- Voice-based AI assistant
- Bangla language support
- AI learning analytics
- Video lecture recommendation system
- Cloud-based file storage

10. CONCLUSION

AI Study Assistant Platform is an AI-integrated educational system designed to improve students' learning experience through intelligent automation and academic assistance. The system combines Artificial Intelligence with modern web technologies to provide chatbot support, quiz generation, study planning, and note summarization features.

The project has strong practical value, future scalability, and real-world educational applications. Through this project, the development team will gain valuable experience in software engineering, Artificial Intelligence integration, full-stack web development, teamwork, and project management.